

Balancing Total Energy Consumption and Mean Makespan in Data Offloading for Space-Air-Ground Integrated Networks

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Abstract—We study the data offloading problem in space-air-ground integrated networks (SAGINs) by jointly optimizing task scheduling and power control to balance the total energy consumption and mean makespan. We consider a mixed integer nonlinear programming problem to minimize a normalized weighted combination of these two conflicting objectives. We first propose an approximation algorithm to find a high-quality solution, which is shown to be at most $\frac{1}{2}$ from the optimum to this problem for given power allocation. We further show that optimal power allocation can be obtained in closed form under the assumption that satellite-ground links have low signal-to-noise ratio (SNR). Thus, the proposed approximation algorithm can be directly utilized to obtain a constant-factor solution to the studied problem in low-SNR scenarios. To extend our solution to more general scenarios, we further propose an efficient hybrid algorithm based on a genetic framework. Our simulation results demonstrate the near-optimality and correctness of the proposed algorithms, and they unveil the interplay between total energy consumption and mean makespan in SAGINs as well.

Index Terms—Space-air-ground integrated networks, data offloading, total energy consumption, mean makespan

1 INTRODUCTION

WITH the rapid development of communication technologies, new commercial applications spring up in large numbers, such as earth monitoring, maritime applications, smart cities, Internet of things, etc [1], [2], [3]. These new applications pose great challenges in both network resources and network coverage. Solely relying on ground networks is difficult to support these applications due to scarce communications resources and limited communications coverage. To complement ground networks, there's a growing recognition in both academia and industry that a large number of

satellites, unmanned aerial vehicles (UAVs), and balloons in space platforms and air platforms, not only improve network performance but also extend network coverage [4]. Thus, the integration of space networks, air networks, and ground networks to construct a space-air-ground integrated network (SAGIN) has become an important way forward [5].

As illustrated in Fig. 1, SAGIN is a hierarchical network where communication nodes at different altitudes complement each other [6]. This multi-layered architecture leads to significant advantages in large-scale coverage, invulnerability, and high throughput. In view of this, SAGINs are widely used in many practical applications, such as environmental monitoring, mapping, meteorology, and natural disaster surveillance [7], [8], [9]. Recently, SAGINs have received a large amount of capital investment. Some commercial companies, such as SpaceX, OneWeb, and O3B, plan to deploy a large number of satellites in space. Moreover, there are many industrial research programs devoted to developing high altitude pseudo-satellite (HAPS) systems, such as Stratobus Platform of Thales, Zephyr Platform of Airbus, etc[10].

However, the fast proliferation of satellites, UAVs, and balloons in SAGINs has brought an explosive growth in space data. Meanwhile, the high-velocity movement of satellites leads to intermittent and limited transmission links [11]. This results in an increasingly prominent conflict between data offloading and transmission resource limitation in SAGINs. Therefore, it is important to investigate how to allocate the limited transmission resources of SAGINs across space, air, and ground segments to efficiently offload the collected data.

Furthermore, compared with stand-alone ground networks, the majority of network nodes in SAGINs are battery-powered and energy-constrained satellites and aircrafts [12].

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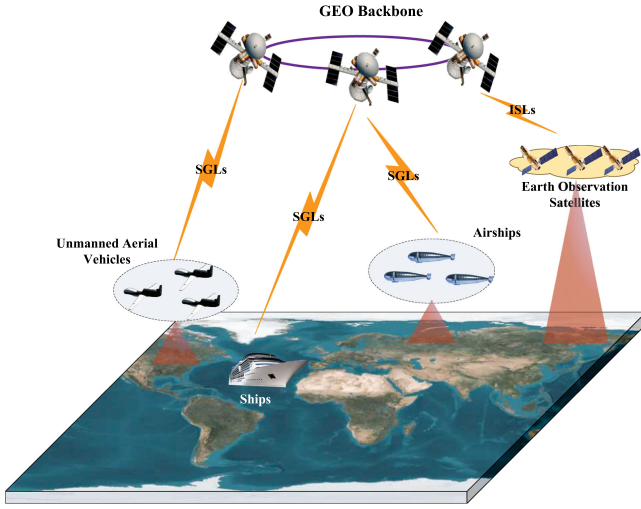


Fig. 1. Space-air-ground integrated networks.

For example, the available energy stored in HAPs is generally insufficient to support overnight use during winter months [13]. Moreover, the energy consumption not only determines the size of but also the lifetime of these battery-carrying satellites and aircrafts [14]. This constitutes a major bottleneck for data offloading in SAGINs. Therefore, it is critical to minimize the energy consumption of each network node to improve the data offloading efficiency of SAGINs. However, focusing only on energy minimization would prolong the makespan of data offloading tasks, which represents the time required for any data offloading task to be completed. In general, the makespan indicates the timeliness of space data and further determines the use value of space data in SAGINs. This because the use value of space data would be reduced or eliminated if it is offloaded too late [15]. As such, it is also a key performance metric of SAGINs.

In brief, it is imperative to design a data offloading scheme to schedule space tasks in SAGINs, to reduce both energy consumption and mean makespan. This problem is challenging in the following three aspects: 1) Intermittent of communications links. The order of scheduling tasks determines the makespan of each task [16]. Since the satellites in SAGINs orbit the earth with high speed, data offloading can occur only in some short and intermittent time windows. Therefore, the makespan of each data offloading task is very sensitive to the scheduling strategy. This makes it difficult to allocate transmission resources within the intermittent time windows for data offloading tasks to minimize their makespan. 2) Heterogeneity of communications links. Different network nodes in SAGINs offload their collected data in different communications links. These communication links are mainly grouped into two types: satellite-ground links (SGLs) and intermittent inter-satellite links (ISLs). The capacity of different communications links follows different channel models. This further complicates the data offloading problem. 3) Multiple criteria for data offloading. Both task scheduling and power control not only determine total energy consumption but also the makespan. It is challenging to design joint task scheduling and power control to achieve a nontrivial tradeoff between total energy consumption and mean makespan in SAGINs.

In this work, we jointly optimize task scheduling and power control while considering intermittent time windows

in SAGINs, aiming to balance the conflicting objectives of total energy consumption and mean makespan. The main contributions of this work are summarized in the following:

- We put forward an optimization framework to design task scheduling and power control and explore the interplay between the total energy consumption and the mean makespan in SAGINs, under the constraints of data offloading and power. We adopt as the performance metric a normalized weighted sum of the total energy consumption and the mean makespan. This results in a mixed integer nonlinear programming (MINLP) problem that well characterizes the main features of data offloading in SAGINs.
- Leveraging the special structure of SAGINs, we propose an approximation algorithm termed the balanced energy and makespan algorithm with fixed power allocation (BEM-FPA) to solve the aforementioned problem when power allocation is given. It is shown to have $\frac{1}{2}$ performance bound for the studied problem for given power allocation. We further derive a closed-form optimal solution to power allocation, thereby enabling BEM-FPA to directly solve our problem. For the scenario of SGLs with general SNR, we propose a two-layer optimization solution termed balanced energy and makespan (BEM). Specifically, in the upper layer, we adopt a genetic framework by optimizing power to minimize the designed performance metric, and in the lower layer, we call the BEM-FPA algorithm given the resultant power allocation.
- Extensive simulations are conducted on SAGINs to demonstrate the following results: 1) For the low-SNR scenarios, direct application of BEM-FPA results in a constant-factor approximation algorithm that is within 2% of optimality in the worst case. 2) For the scenarios with moderate to high-SNR SGLs, we observe how BEM effectively trades off the conflicting objectives of total energy consumption and mean makespan over a wide range of weights between them. 3) Our proposed algorithm outperforms state-of-the-art alternatives under various parameter settings.

The rest of this paper is organized as follows. Section 2 provides a review of the related work. In Section 3, we present the system model and the problem formulation. In Section 4, we present the efficient data offloading algorithms to solve the studied problem. Section 5 presents simulation results and discussions. Finally, we give concluding remarks in Section 6.

2 RELATED WORK

Many existing works on the data offloading issue for space networks aim to efficiently allocate transmission resources within intermittent time windows. Their data offloading designs are based on the mathematical tools including graph theory [17], [18], [19], metaheuristics [20], [21], [22], convex optimization [23], and game theory [24].

The high dynamics of satellite networks poses new challenges to the design of data offloading. The works of [17],

[18], [19] used graph models to characterize the dynamics of satellite networks. Liu et al. [17] used a time-expanded graph model to study the impact of antenna slewing time on data offloading. Zhou et al. [18] exploited the time-expanded graph to address data offloading with consideration for time-varying and differentiated communication links in data relay satellite networks (DRSNs). Jia et al. [19] used inter-satellite links to enable data offloading among satellites and then employed a space-time topology graph to depict the dynamic topologies of low earth orbit (LEO) satellite networks, aiming to simplify the complexity of the proposed data offloading scheme. However, the construction of graph models in [17], [18], [19] would result in a high computation cost, especially as the number of satellites increases.

To accommodate the large-scale networks of satellites, the works of [20], [21], [22] turned to metaheuristics. Deng et al. [20] proposed an improved genetic task scheduling method to deal with uncertain factors in DRSNs. To address the problem of data breakpoint transmission in DRSNs, Chen et al. [21] developed a task scheduling method based on the variable neighborhood descent method. Furthermore, Dai et al. [22] devised a dynamic task scheduling method based on an improved genetic algorithm, aiming to ensure the data offloading efficiency of emergency tasks in DRSNs.

In addition to metaheuristics, our prior work [23] utilized a stochastic optimization method to perform dynamic hybrid task scheduling for DRSNs, aiming to maximize the number of hybrid tasks. By using a three-sided matching mechanism, Jia et al. [24] cooperated the transmission resources across LEO satellites and HAPs to maximize the revenue of SAGINs.

The aforementioned studies have considered some practical influencing factors into their data offloading models, such as antenna slewing time [17], time-varying and differentiated communication links [18], inter-satellite links [19], randomness of task arrivals [20], [23], randomness of system environment [21], emergency tasks [22], and resources cooperation [24]. However, these works do not consider energy consumption, which could lead to shortening of the lifetime of satellites and aircrafts, thereby deteriorating the data offloading performance of SAGINs.

Recent works have addressed the energy consumption issues in SAGINs. Dong et al. [25] proposed an integrated HAP-Satellite architecture to support a novel energy-efficient transmission scheme for emergency scenarios. By optimizing the ratio of average rate over total power consumption, An et al. [26] proposed an energy-efficient power allocation mechanism for land mobile satellite systems. Ji et al. [27] jointly optimized power control and downlink resources to improve the energy efficiency of data offloading for multi-cell satellite-terrestrial networks. With the consideration of energy consumption, time-varying links, and the differentiation for tasks, Zhou et al. [28] proposed an efficient data offloading strategy for small satellite networks.

However, it is difficult to directly apply the above works to solve the data offloading problem for SAGINs due to the following reasons: First, the works [25], [26], [27] neglect the dynamics topologies in their schemes. Second, the energy

consumption in [28] is characterized as resource constraints in their problem model of data offloading, which does not allow direct tuning of energy efficiency. Third, none of [25], [26], [27], [28] study the impact of heterogeneity of communications links on the data offloading efficiency of SAGINs. Finally, the works [26], [27], [28] address the data offloading problem only in terms of minimizing energy consumption, which could prolong the makespan of tasks and further decrease the quality of data offloading.

In contrast to the above, in this work we study joint task scheduling and power control to reduce the energy consumption and makespan at the same time, both of which are important to the data offloading efficiency of SAGINs. Interestingly, it is found that the joint task offloading and power control problems have been well investigated in the context of edge computing networks. Specifically, Guo et al. [29] devised online offloading strategies to minimize the average task execution delay with considering the constraint of average energy consumption. Vu et al. [30] proposed an improved branch-and-bound algorithm to minimize the total energy consumption under the constraints of service delay. However, these offloading strategies are very hard to solve our problem. This is because data offloading in edge computing networks can occur in the whole scheduling horizon, while data offloading in SAGINs can occur only in some short and intermittent time windows. Hence, to address the data offloading in SAGINs with high efficiency, it is necessary to consider the special property of the SAGINs as our work.

3 SYSTEM MODEL AND PROBLEM FORMULATION

We consider an SAGIN as shown in Fig. 1. Notations used in the following are listed in Table 1. We divide the scheduling horizon of the SAGIN into a set of fixed-size time slots, denoted by $\mathcal{T} = \{1, \dots, T\}$. Let time slot $t \in \mathcal{T}$ index the time interval of $[t, t+1]$. In the SAGIN, the data collectors are denoted by set $\mathcal{S} = \{1, \dots, S\}$, which consists of spaceborne (e.g., low-orbit earth observation satellites), airborne (e.g., UAVs, airships), and terrestrial (e.g., ships) nodes. The data transmission tasks of the data collectors are denoted by set $\mathcal{I} = \{1, \dots, I\}$, which can be offloaded through either SGLs or ISLs. We let $s(i)$ index the data collector collecting the data of task i . The data collectors transmit their collected data to data sinks, which are a set of data relay satellites (DRSs). For ease of analysis processing, we do not consider ground stations in this scenario. We let $\mathcal{H} = \{1, \dots, H\}$ denote the set of data receiver antennas on the DRSs. For clarify, we further divide set $\mathcal{H}$ into $\mathcal{H}_1$ and $\mathcal{H}_2$, i.e., $\mathcal{H}_1 \cup \mathcal{H}_2 = \mathcal{H}$, such that $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively denote the set of data receiver antennas of DRSs to access non-spacecrafts and spacecrafts (i.e., low-orbit satellites). Let $h(s)$ be the data receiver antenna accessed by data collector s . Finally, we define $\mathcal{I}_1 = \{i | h(s(i)) \in \mathcal{H}_1\}$ and $\mathcal{I}_2 = \{i | h(s(i)) \in \mathcal{H}_2\}$ as the set of data transmission tasks from non-spacecraft and spacecraft, respectively, such that $\mathcal{I}_1 \cup \mathcal{I}_2 = \mathcal{I}$.

An ISL or SGL is generated only when data collector $s \in \mathcal{S}$ moves within the coverage of antenna $h \in \mathcal{H}$ on a DRS. Particularly, the relative movement between data collectors and data sinks leads to intermittent ISLs and SGLs. We term the activation of each ISL or SGL as the transmission time

TABLE 1
Notation List

Notation	Description
i, j	task index
T	maximum time slot index
$\mathcal{T}$	set of fixed-size time slots (i.e., $\mathcal{T} = \{1, \dots, T\}$)
S	maximum data collector index
$\mathcal{S}$	set of data collectors (i.e., $\mathcal{S} = \{1, \dots, S\}$)
H	maximum data receiver antenna index
$\mathcal{H}$	set of data receiver antennas (i.e., $\mathcal{H} = \{1, \dots, H\}$)
$h(s)$	data receiver antenna accessed by data collector s
$\mathcal{K}_{s,h}$	set of TTWs between data collectors s and antenna h
$\mathcal{K}_{S,h}$	set of TTWs between all data collectors and antenna h
$G_{s(i)}^{\text{tran}}$	transmit antenna gain of data collector $s(i)$
G_h^{rec}	gain of data receiver antenna h
L_f	free space loss
L_l	total line loss
N	noise power
B_c	bandwidth
R_{ih}^{SGL}	achievable data rate of SGLs
κ	Boltzmann's constant (in JK^{-1})
T_s	total system noise temperature (in K)
$(E_b/N_0)_{\text{req}}$	required ratio of received energy-per-bit to noise-density
M	link margin
R_{ih}^{ISL}	achievable data rate of ISLs
p_{ih}	transmission time of task i on antenna h
D_i	data size of task i
a_k	beginning time of TTW k
b_k	end time of TTW k
$\mathcal{F}(i, h)$	set of feasibility time slots for task i within the TTWs on any antenna h
$\mathcal{F}(i)$	set of all the feasibility time slots for task i
$\Phi(i, h, t)$	set of occupying time slots for task i to start offloading in time slot t on antenna h
$P_{\min}^{s(i)}$	minimum transmit power of data collector $s(i)$
$P_{\max}^{s(i)}$	maximum transmit power of data collector $s(i)$
E_i	energy consumption to transmit the data of task i
$E_{\max}$	maximum total energy consumption of all data collectors
E_{tol}^*	optimal total energy consumption of all data collectors
C_i	completion time to offload the data of task i
$C_{\max}$	maximum mean makespan
C^*	optimal mean makespan
λ	weighting factor

window (TTW). In general, the periodic motion of data collectors s leads to multiple TTWs associated with antenna h , denoted by $\mathcal{K}_{s,h}$. We further introduce a two-tuple (a_k, b_k) to depict the TTW $k \in \mathcal{K}_{s,h}$ with a_k and b_k respectively representing its beginning and end times. Let $\mathcal{K}_{S,h} = \bigcup_{s \in \mathcal{S}} \mathcal{K}_{s,h}$.

In this work, we aim to study the data offloading problem that how to efficiently offload the space data from a set of DRSs to data collectors. As such, we only consider the data offloading occurring on the one-hop ISLs between DRSs and data collectors without consideration of the data offloading among LEO satellites. Moreover, the data offloading between data collectors and ground users is out of the scope of this work.

3.1 Channel Model

In this subsection, we present the channel models for SGLs and ISLs as follows.

3.1.1 SGLs

We denote by P_i the transmit power (in W) allocated to task i . Let $\mathbf{P} = \{P_i\}$. The SNR of SGLs can be evaluated by

$$\text{SNR} = \frac{P_i G_{s(i)}^{\text{tran}} G_h^{\text{rec}} L_f L_l}{N}, \forall i \in \mathcal{I}, h \in \mathcal{H}_1, \quad (1)$$

where $G_{s(i)}^{\text{tran}}$ is the transmit antenna gain of data collector $s(i)$, G_h^{rec} is the gain of data receiver antenna h , L_f is the free space loss, L_l is the total line loss, and N is the noise power. We use the Shannon formula to compute the achievable data rate of SGLs

$$R_{ih}^{\text{SGL}} = B_c \log_2(1 + \text{SNR}), \forall i \in \mathcal{I}, h \in \mathcal{H}_1, \quad (2)$$

with B_c being the bandwidth. Here, we assume that data offloading on SGLs is error-free transmission and some impact factors on SGLs (e.g., rain attenuation, cloud, fog) can be dealt with transmission schemes in physical layers.

3.1.2 ISLs

The achievable data rate on ISLs [31], [32], [33] is expressed as

$$R_{ih}^{\text{ISL}} = \frac{P_i G_{s(i)}^{\text{tran}} G_h^{\text{rec}} L_f L_l}{\kappa T_s (E_b/N_0)_{\text{req}} M}, \forall i \in \mathcal{I}, h \in \mathcal{H}_2, \quad (3)$$

where κ is the Boltzmann's constant (in JK^{-1}), T_s is the total system noise temperature (in K), $(E_b/N_0)_{\text{req}}$ is the required ratio of received energy-per-bit to noise-density, and M is the link margin.

3.2 Constraints

We first introduce binary variable $x_{iht} \in \{0, 1\}$ to represent the offloading strategy, such that $x_{iht} = 1$ indicates that the data of task i is started to be offloaded in time slot $t \in \mathcal{T}$ on antenna h ; otherwise $x_{iht} = 0$. Let $\mathbf{x} = \{x_{iht}\}$. We define a feasible time slot within TTW k for task i as the time slot in which the data of task i can be started to be offloaded. As shown in Fig. 2, the feasible time slots for task i within TTW k are t_3, t_4, t_5 , and t_6 . As such, we can define the set of feasibility time slots for task i within the TTWs on any antenna h as

$$\mathcal{F}(i, h) = \bigcup_{k \in \mathcal{K}_{s(i),h}} [a_k, b_k - p_{ih}]. \quad (4)$$

Wherein, p_{ih} represents the transmission time of task i on antenna h , which is calculated by

$$p_{ih} = \frac{D_i}{R_{ih}^{\text{SGL}}}, \forall h \in \mathcal{H}_1, \quad (5)$$

and

$$p_{ih} = \frac{D_i}{R_{ih}^{\text{ISL}}}, \forall h \in \mathcal{H}_2, \quad (6)$$

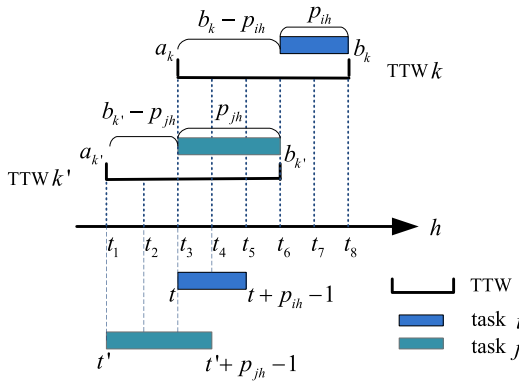


Fig. 2. Illustration for the sets of feasibility time slots and occupying time slots.

with D_i being the data size of task i . As such, we can introduce the following constraint to state that the data of each task is offloaded exactly once within the feasibility time slots on all the antennas

$$\sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht} = 1, \forall i \in \mathcal{I}. \quad (7)$$

Next, we let $\mathcal{F}(i) = \bigcup_{h \in \mathcal{H}} \mathcal{F}(i, h)$ denote the set of all the feasibility time slots for task i . As such, we have the binary constraints as follows:

$$x_{iht} \in \{0, 1\}, \forall i \in \mathcal{I}, t \in \mathcal{F}(i), \quad (8)$$

$$x_{iht} = 0, \forall i \in \mathcal{I}, t \notin \mathcal{F}(i). \quad (9)$$

We further observe from Fig. 2 that the data offloading of task i occupies the feasible time slots within $[t, t + p_{ih} - 1]$ with t being the time slot to start offloading. We define the set of occupying time slots for task i to start offloading in time slot t on antenna h as

$$\Phi(i, h, t) = [t, t + p_{ih} - 1] \cap \mathcal{F}(i, h). \quad (10)$$

Furthermore, we use binary variables ρ_{iht} to represent time slot allocation, so that $\rho_{iht} = 1$ indicates that time slot t of antenna h is occupied to offload the data of task i ; otherwise $\rho_{iht} = 0$. Let $\boldsymbol{\rho} = \{\rho_{iht}\}$. As such, we introduce the following three constraints to implicitly represent the *antenna capacity*: each antenna can transmit the data of at most one task in any time slot

$$\sum_{i \in \mathcal{I}} \rho_{iht} \leq 1, \forall h \in \mathcal{H}, t \in \mathcal{T}, \quad (11)$$

$$\rho_{ihn} \geq x_{iht}, \forall n \in \Phi(i, h, t), i \in \mathcal{I}, h \in \mathcal{H}, t \in \mathcal{T}, \quad (12)$$

$$\rho_{iht} \in \{0, 1\}, \forall i \in \mathcal{I}, h \in \mathcal{H}, t \in \mathcal{T}. \quad (13)$$

Finally, we introduce the power control constraint as follows:

$$P_{\min}^{s(i)} \leq P_i \leq P_{\max}^{s(i)}, \forall i \in \mathcal{I}, \quad (14)$$

where $P_{\min}^{s(i)}$ and $P_{\max}^{s(i)}$ respectively denote the minimum transmit power and the maximum transmit power of data collector $s(i)$.

3.3 Problem Formulation

We denote by E_i the energy consumption to transmit the data of task i , which is given by

$$E_i = \sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht} P_i p_{ih}. \quad (15)$$

Then, the total energy consumption of the data collectors is

$$E_{\text{tol}} = \sum_{i \in \mathcal{I}} E_i = \sum_{i \in \mathcal{I}} \sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht} P_i p_{ih}. \quad (16)$$

Let $E_{\max}$ and E_{tol}^* be the maximum total energy consumption and the optimal total energy consumption of all data collectors, respectively.

Let C_i be the completion time to offload the data of task i . We have

$$C_i = \sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht} (t + p_{ih} - 1). \quad (17)$$

We define the mean makespan as the mean sum of the completion time of tasks, which is expressed as $C = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} C_i$. Let $C_{\max}$ and C^* be the maximum mean makespan and optimal mean makespan, respectively.

We focus on a static data offloading problem to minimize the total energy consumption and the mean makespan of tasks. Here, we adopt the normalized weighted sum method [34] to balance the two conflicting objectives of minimizing both total energy consumption and mean makespan. This can provide a complete Pareto-optimal set [35] for our problems. In the normalized weighted sum method, each objective function should be first normalized in $(0, 1)$ aiming to give equal emphasis, and then multiplied by weights. Here, we introduce a weighting factor $\lambda \in [0, 1]$ to balance two designed objectives. In practical applications, we tune the values of λ in $[0, 1]$ to obtain a set of mean makespan and total energy consumption for choice. Thus, we consider an MINLP problem given by

$$\begin{aligned} \mathbf{P0}: \min_{\mathbf{x}, \mathbf{P}} & \left(\lambda \frac{C - C^*}{C_{\max} - C^*} + (1 - \lambda) \frac{E_{\text{tol}} - E_{\text{tol}}^*}{E_{\max} - E_{\text{tol}}^*} \right) \\ \text{s.t.} & (7) - (9), \\ & (11) - (14). \end{aligned}$$

Note that compared with aerial networks or ground networks, one key characteristic of SAGINs is in the intermittent ISLs due to the high-speed motion of LEO satellites. In particular, the intermittent ISLs are depicted as the time window constraints of (7), (11), and (12) in **P0**. As such, **P0** is capable of capturing the key characteristic of SAGINs.

3.4 Problem Analysis

For analytical tractability, we rewrite **P0** in the following form

$$\begin{aligned} \mathbf{P0}: \min_{\mathbf{x}, \mathbf{P}} & \left(\lambda \frac{C}{C_{\max} - C^*} + (1 - \lambda) \frac{E_{\text{tol}}}{E_{\max} - E_{\text{tol}}^*} \right) - \mathcal{U} \\ \text{s.t.} & (7) - (9), \\ & (11) - (14), \end{aligned}$$

where a constant $\mathcal{U}$ is defined as

$$\mathcal{U} = \lambda \frac{C^*}{C_{\max} - C^*} + (1 - \lambda) \frac{E_{\text{tol}}^*}{E_{\max} - E_{\text{tol}}^*}.$$

The following two preliminary observations reveal the different properties of the studied problem depending on whether $\lambda > 0$. It is obvious that when $\lambda = 0$, $\mathbf{P0}$ degrades into an energy minimization problem. In this case, since a smaller transmission power leads to a longer transmission duration, which can increase the energy consumption. In general, it is non-trivial to find the optimal power to minimize the energy consumption. However, as shown in the theorem below, in our problem, the minimum transmit power is optimal.

Theorem 1. When $\lambda = 0$, we have $P_i^* = P_{\min}^{s(i)}, \forall i$.

Proof. See Appendix A, (available online). $\square$

Theorem 2. $\mathbf{P0}$ is NP-hard in general when $\lambda \in (0, 1]$.

Proof. See Appendix B, available in the online supplemental material. $\square$

4 PROPOSED SOLUTIONS

In this section, we first devise an approximation algorithm, termed BEM-FPA, with performance guarantee to solve $\mathbf{P0}$ for given power allocation. Then, utilizing this algorithm as a building block, we solve $\mathbf{P0}$ for the scenarios with both low-SNR SGLs and more general SGLs.

4.1 Approximation Algorithm With Performance Guarantee for Given Power Allocation

With given power allocation, $\mathbf{P0}$ becomes the following problem

$$\begin{aligned} \mathbf{P0}^{\text{fixed power}}: \min_{\mathbf{x}} & \mathcal{G} \sum_{i \in \mathcal{I}} \sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht}(t + p_{ih} - 1) + \mathcal{V} \\ \text{s.t.} & (7) - (9), \\ & (11) - (13), \end{aligned}$$

where constants $\mathcal{G}$ and $\mathcal{V}$ are respectively defined as

$$\mathcal{G} = \frac{\lambda}{|\mathcal{I}|(C_{\max} - C^*)}$$

and

$$\mathcal{V} = (1 - \lambda) \frac{E_{\text{tol}} - E_{\text{tol}}^*}{E_{\max} - E_{\text{tol}}^*} - \mathcal{G}C^*.$$

For the convenience of analysis, we remove the constant terms $\mathcal{G}$ and $\mathcal{V}$ from $\mathbf{P0}^{\text{fixed power}}$ without changing its optimal solutions, to obtain the following task scheduling optimization problem

$$\begin{aligned} \mathbf{P1}: \min_{\mathbf{x}} & \sum_{i \in \mathcal{I}} \sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht}(t + p_{ih} - 1) \\ \text{s.t.} & (7) - (9), \\ & (11) - (13). \end{aligned}$$

To solve $\mathbf{P0}^{\text{fixed power}}$ efficiently, we first design a BEM-FPA algorithm with performance guarantee to solve $\mathbf{P1}$ in Section 4.1.1, and then determine the values of parameters $\mathcal{G}$ and $\mathcal{V}$ by evaluating the values of E_{tol}^* , $E_{\max}$, C^* , and $C_{\max}$ in Section 4.1.2.

4.1.1 Task Scheduling Optimization

It would be easy to obtain heuristic solution to $\mathbf{P1}$ through the Lagrangian Fix-and-Relax method developed in [36]. However, it is hard to quantify the performance gap between such a solution and the optimal solution. Therefore, we do not use this method to solve $\mathbf{P1}$ but turn to leverage the convex quadratic and semidefinite relaxation (CQSR) method [37] to transform $\mathbf{P1}$ into a convex quadratic program. Then, we use the rounding method PIPAGE [38] to convert the fractional solution to this convex quadratic program into an integral solution. Finally, we prove that the obtained integral solution has a constant-factor performance guarantee.

We observe that the difficulty to solve $\mathbf{P1}$ is in the time window constraints of (7), (11), and (12). In what follows, we first exploit the special characteristics of space network to tackle the time window constraints aiming at simplifying $\mathbf{P1}$, such that the CQSR method can be used to solve it. In particular, we observe that the considered network is consist of DRSS, which are in geosynchronous earth orbit (GEO). Also, three such DRSSs are capable of providing 100 percent global coverage. As such, several DRSSs are concentrated in three locations instead of uniformly distributed in GEO. This special structure of satellite constellation is used in the practical data relay satellite systems such as tracking and data relay satellite system (TDRSS) [39], Tianlian system [40], and European data relay satellite (EDRS) system [41]. Since the transmit antennas on DRSSs in the same location are close to each other, they may be regarded as having the same TTWs.

Notice that the endpoints of TTWs $\mathcal{K}_{S,h}$ partition the scheduling horizon of any antenna h into a set of time segments $(t_l, t_{l+1}), l = 1, 2, \dots$, as shown in Fig. 3a. For any antenna h , we merge these time segments into time-slot blocks by calculating the union set of $\mathcal{K}_{S,h}$, as shown Fig. 3b. For example, the union set of time windows (t_1, t_3) and (t_2, t_4) in Fig. 3a is the time-slot block (t_1, t_4) in Fig. 3b. As such, we merge time segments $(t_l, t_{l+1}), l = 1, 2, 3$ into time-slot block (t_1, t_4) . Then, we need to construct a compressed space by compressing the time axis to merge the produced time-slot blocks, as shown Fig. 3c. As such, the data

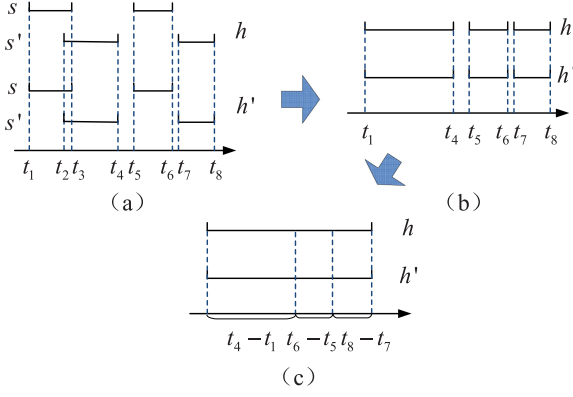


Fig. 3. Time block construction.

offloading problem in the compressed space is without the time window constraints, thereby significantly simplifying **P1**.

We refer to each merged time-slot block as a machine and let $\mathcal{M} = \{1, \dots, M\}$ denote the set of machines. Denote by Δ_{im} the transmission time of task i on machine m . In what follows, our main idea is that each task i is first assigned to a machine, followed by which the tasks are sequenced in each machine. It is clear that the above two steps are without losing optimality of **P1**. We introduce new variables $y_{im} \in \{0, 1\}$ to present the scheduling strategy, where $y_{im} = 1$ indicates that task i is allocated to antenna m to offload its data; otherwise $y_{im} = 0$. Furthermore, we denote by $\prec_m$ an order for any machine m on $\mathcal{I}$ to sequence tasks. In particular, it is an optimal assignment of tasks to each machine according to the Smith's rule [16]: $j \prec_m i$ if $\Delta_{jm} \leq \Delta_{im}, \forall m \in \mathcal{M}$. As such, we give a reformulation of **P1** over a compressed time space as follows:

$$\begin{aligned} \mathbf{P2}: \quad & \min_{\mathbf{y}} \sum_{i \in \mathcal{I}} C'_i \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \end{aligned} \quad (18)$$

$$C'_i = \sum_{m \in \mathcal{M}} y_{im} (\Delta_{im} + \sum_{j \prec_m i} y_{jm} \Delta_{jm}), \forall i \in \mathcal{I}, \quad (19)$$

$$y_{im} \in \{0, 1\}, \forall i \in \mathcal{I}, m \in \mathcal{M}. \quad (20)$$

Constraints (18) reflect that each task is assigned to exactly one machine. Constraints (19) is the completion time. Notice that C'_i in the objective function can be replaced by the term on the right-hand side of constraints (19), thus getting rid of constraints (19). Constraints (20) are binary constraints.

Then, we further remove constraints (19) from **P2** to obtain the following form

$$\begin{aligned} \mathbf{P3}: \quad & \min_{\mathbf{y}} \sum_{i \in \mathcal{I}} \sum_{m \in \mathcal{M}} y_{im} (\Delta_{im} + \sum_{j \prec_m i} y_{jm} \Delta_{jm}) \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & y_{im} \in \{0, 1\}, \forall i \in \mathcal{I}, m \in \mathcal{M}. \end{aligned}$$

In the following, we define $\mathbf{c}$ as a vector of length IM given by $c_{im} = p_{im}$. Let $W = (w_{(im)(jm')})$ be a symmetric $IM \times IM$ -matrix with

$$w_{(im)(jm')} = \begin{cases} 0 & m \neq m' \text{ or } i = j, \\ p_{im} & m = m' \text{ and } i \prec_m j, \\ p_{jm} & m = m' \text{ and } j \prec_m i. \end{cases} \quad (21)$$

It is observed from (21) that we can decompose W into $|\mathcal{M}|$ diagonal blocks, each corresponding to a time-slot block of a machine m , denoted by $W_m, \forall m \in \mathcal{M}$

$$W_m = \begin{pmatrix} 0 & \Delta_{1m} & \Delta_{1m} & \cdots & \Delta_{1m} \\ \Delta_{1m} & 0 & \Delta_{2m} & \cdots & \Delta_{2m} \\ \Delta_{1m} & \Delta_{2m} & 0 & & \Delta_{3m} \\ \vdots & \vdots & & \ddots & \vdots \\ \Delta_{1m} & \Delta_{2m} & \Delta_{3m} & \cdots & 0 \end{pmatrix}. \quad (22)$$

Then, **P3** can be written in the following form

$$\begin{aligned} \mathbf{P4}: \quad & \min_{\mathbf{y}} F(\mathbf{y}) = \mathbf{c}^T \mathbf{y} + \frac{1}{2} \mathbf{y}^T W \mathbf{y} \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & y_{im} \in \{0, 1\}, \forall i \in \mathcal{I}, m \in \mathcal{M}. \end{aligned}$$

We relax $\mathbf{y}$ to be continuous and obtain the following problem

$$\begin{aligned} \mathbf{P5}: \quad & \min_{\mathbf{y}} F(\mathbf{y}) = \mathbf{c}^T \mathbf{y} + \frac{1}{2} \mathbf{y}^T W \mathbf{y} \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & 0 \leq y_{im} \leq 1, \forall i \in \mathcal{I}, m \in \mathcal{M}. \end{aligned} \quad (23)$$

The optimal objective value to **P5** is equal to that of **P4**, which can be easily proved according to the corollary 2.3 in [37]. Furthermore, one can observe that **P5** is convex if and only if matrix W is positive semidefinite. Unfortunately, matrix W is not always positive semidefinite. To this end, we need to raise the diagonal entries of W , such that it is positive semidefinite. This leads to the following new optimization problem, given by

$$\begin{aligned} \mathbf{P6}: \quad & \min_{\mathbf{y}} G_{\xi}(\mathbf{y}) = (1 - \xi) \mathbf{c}^T \mathbf{y} + \frac{1}{2} \mathbf{y}^T (W + 2\xi \text{diag}(\mathbf{c})) \mathbf{y} \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & 0 \leq y_{im} \leq 1, \forall i \in \mathcal{I}, m \in \mathcal{M}, \end{aligned}$$

where ξ is a positive constant within $(0, 1)$.

To reveal the relationship between **P5** and **P6**, we give the following theorem:

Theorem 3. *P6 provides a lower bound of the optimal objective value to P5 for any $\xi > 0$.*

Proof. See Appendix C, available in the online supplemental material. $\square$

Inspired by Theorem 3, we aim to find a tight lower bound of **P5**. Therefore, we consider the following optimization problem

$$\begin{aligned} \mathbf{P7}: \quad & \min_{\mathbf{y}} \max_{\xi} G_{\xi}(\mathbf{y}) \\ \text{s.t.} \quad & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & 0 \leq y_{im} \leq 1, \forall i \in \mathcal{I}, m \in \mathcal{M}. \end{aligned}$$

Obviously, **P7** can be rewritten as the following form

$$\begin{aligned} \mathbf{P8}: \quad & \min_{\mathbf{y}, Z} Z \\ \text{s.t.} \quad & Z \geq G_{\xi}(\mathbf{y}), \forall \xi, \\ & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & 0 \leq y_{im} \leq 1, \forall i \in \mathcal{I}, m \in \mathcal{M}, \\ & Z \geq 0. \end{aligned}$$

We use the following lemma that specifies when **P8** is a convex problem.

Lemma 1 (Lemma 2.4, [37]). *P8 is convex if and only if $\xi \geq \frac{1}{2}$.*

Then, the following theorem reveals the relationship between the optimal objective values of **P5** and **P8**.

Theorem 4. *The optimal objective value to **P8** is 2 times the optimal objective value to **P5** if $\xi = \frac{1}{2}$.*

Proof. See Appendix D, available in the online supplemental material. $\square$

Thus, we set $\xi = \frac{1}{2}$ to rewrite **P8** as the following problem

$$\begin{aligned} \mathbf{P9}: \quad & \min_{\mathbf{y}, Z} Z \\ \text{s.t.} \quad & Z \geq G_{\frac{1}{2}}(\mathbf{y}) = \frac{1}{2} \mathbf{c}^T \mathbf{y} + \frac{1}{2} \mathbf{y}^T (W + \text{diag}(\mathbf{c})) \mathbf{y}, \\ & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & 0 \leq y_{im} \leq 1, \forall i \in \mathcal{I}, m \in \mathcal{M}, \\ & Z \geq 0. \end{aligned}$$

Furthermore, we substitute $\xi = \frac{1}{2}$ to (34) to yield

$$\begin{aligned} F(\mathbf{y}) &= G_{\frac{1}{2}}(\mathbf{y}) + \frac{1}{2} (\mathbf{c}^T \mathbf{y} - \mathbf{y}^T \text{diag}(\mathbf{c}) \mathbf{y}) \\ &\leq Z + \frac{1}{2} (\mathbf{c}^T \mathbf{y} - \mathbf{y}^T \text{diag}(\mathbf{c}) \mathbf{y}). \end{aligned} \quad (24)$$

It is observed from (24) that if $Z \geq \mathbf{c}^T \mathbf{y}$, then we obtain

$$F(\mathbf{y}) \leq \frac{3}{2} Z - \frac{1}{2} \mathbf{y}^T \text{diag}(\mathbf{c}) \mathbf{y} \leq \frac{3}{2} Z. \quad (25)$$

This means that we can add a lower bound constraint on Z (i.e., $Z \geq \mathbf{c}^T \mathbf{y}$) to improve the approximation of **P9**, as shown in Theorem 5 below. This leads to a relaxed problem as follows:

$$\begin{aligned} \mathbf{P10}: \quad & \min_{\mathbf{y}, Z} Z \\ \text{s.t.} \quad & Z \geq \frac{1}{2} \mathbf{c}^T \mathbf{y} + \frac{1}{2} \mathbf{y}^T (W + \text{diag}(\mathbf{c})) \mathbf{y}, \\ & Z \geq \mathbf{c}^T \mathbf{y}, \\ & \sum_{m \in \mathcal{M}} y_{im} = 1, \forall i \in \mathcal{I}, \\ & 0 \leq y_{im} \leq 1, \forall i \in \mathcal{I}, m \in \mathcal{M}, \\ & Z \geq 0. \end{aligned}$$

We propose to solve **P0**^{fixed power} by first solving **P10** and then using the PIPAGE method [38] to recover an integer solution. This algorithm is termed BEM-FPA and is summarized in Algorithm 1.

Algorithm 1. Balanced Energy and Makespan Algorithm With Fixed Power Allocation (BEM-FPA)

Input: Given $\xi = \frac{1}{2}$ and $P_i, \forall i$.

1: Formulate **P10**.

2: Obtain an optimal solution $\mathbf{y}^*$ to **P10**.

3: Use the PIPAGE method to convert $\mathbf{y}^*$ into integer solution $\bar{\mathbf{y}}$.

Output: $\bar{\mathbf{y}}$.

Furthermore, the following result reveals the approximation performance of BEM-FPA.

Theorem 5. *BEM-FPA produces a $\frac{1}{2}$ -approximate solution to **P0**^{fixed power}.*

Proof. See Appendix E, available in the online supplemental material. $\square$

Theorem 6. *BEM-FPA has a time complexity of $\mathcal{O}((IM)^{4.5} \log(1/\epsilon))$, where $\epsilon \geq 0$ is the solution accuracy.*

Proof. See Appendix F, available in the online supplemental material. $\square$

4.1.2 Parameters Determination

To obtain the values of $\mathcal{G}$ and $\mathcal{V}$, we need to determine the values of E_{tol}^* , E_{max}^* , C^* , and C_{max} . We note that the key idea of the normalized weighted sum method [34] is to transform multiple objective functions with different range and units into their normalized weighted sum (i.e., the sum of non-dimensional objective functions), while guaranteeing that each normalized objective function is a function with a lower limit of zero and an upper limit of one. Theorem 2 indicates that computing the optimal values of E_{tol}^* and C^* could result in a prohibitive cost. To reduce computational cost, the authors of [34] propose a suggestion that one can use their approximations for each optimal value of objective function to guarantee that each designed objective has a value between zero and one. Along this idea, we adopt the lower bounds of E_{tol}^* and C^* as their approximations, denoted by $\hat{E}_{\text{tol}}^*$ and $\hat{C}^*$, respectively.

The following theorem indicates the values of $\hat{E}_{\text{tol}}^*$ and E_{max}^* .

Theorem 7. The lower bound and upper bound of E_{tol}^* are respectively as follows:

$$\begin{aligned} \hat{E}_{\text{tol}}^* &= \sum_{i \in \mathcal{I}_1} \frac{P_{\min}^{s(i)} D_i}{B_c \log \left(1 + \frac{P_{\min}^{s(i)} G_{s(i)}^{\text{tran}} G_{h_1^*}^{\text{rec}} L_f L_l}{N} \right)} \\ &\quad + \sum_{i \in \mathcal{I}_2} \frac{D_i \kappa T_s (E_b/N_0)_{\text{req}} M}{G_{s(i)}^{\text{tran}} G_{h_2^*}^{\text{rec}} L_f L_l}, \\ h_1^* &= \operatorname{argmax}\{G_h^{\text{rec}}, h \in \mathcal{H}_1\}, \\ h_2^* &= \operatorname{argmax}\{G_h^{\text{rec}}, h \in \mathcal{H}_2\}, \end{aligned}$$

and

$$\begin{aligned} E_{\text{max}} &= \sum_{i \in \mathcal{I}_1} \frac{P_{\max}^{s(i)} D_i}{B_c \log \left(1 + \frac{P_{\max}^{s(i)} G_{s(i)}^{\text{tran}} G_{h_3^*}^{\text{rec}} L_f L_l}{N} \right)} \\ &\quad + \sum_{i \in \mathcal{I}_2} \frac{D_i \kappa T_s (E_b/N_0)_{\text{req}} M}{G_{s(i)}^{\text{tran}} G_{h_4^*}^{\text{rec}} L_f L_l}, \\ h_3^* &= \operatorname{argmin}\{G_h^{\text{rec}}, h \in \mathcal{H}_1\}, \\ h_4^* &= \operatorname{argmin}\{G_h^{\text{rec}}, h \in \mathcal{H}_2\}. \end{aligned}$$

Proof. See Appendix G, available in the online supplemental material. $\square$

Furthermore, we sequence tasks on antenna h in order of descending processing time, denoted by $\prec_h$. We give the following theorem:

Theorem 8. The lower bound of C^* is $\hat{C}^* = Z^*$, where Z^* is an optimal solution to **P10** with $P_i = P_{\max}^{s(i)}, \forall i$. The upper bound of C^* is as follows

$$C_{\text{max}} = \sum_{i \in \mathcal{I}_1} \left(p_{ih_3^*} + \sum_{j \prec_{h_3^*} i} p_{jh_3^*} \right) + \sum_{i \in \mathcal{I}_2} \left(p_{ih_4^*} + \sum_{j \prec_{h_4^*} i} p_{jh_4^*} \right).$$

Proof. See Appendix H, available in the online supplemental material. $\square$

As such, we can determine the values of $\mathcal{G}$ and $\mathcal{V}$ according to the values of $\hat{E}_{\text{tol}}^*$, E_{max} , $\hat{C}^*$, and C_{max} .

4.2 Joint Power and Task Scheduling Optimization

4.2.1 Approximation Algorithm with Performance

Guarantee for Low-SNR SGLs

The SAGIN is a kind of digital communication systems, and thus may operate in the power-limited regime (i.e., $\text{SNR} \ll 1$) [42]. As such, in this subsection we aim to obtain an approximation solution to **P0** with performance guarantee for the scenarios of low-SNR SGLs, i.e., $\text{SNR} \ll 1$, where SNR is defined as (1). Specifically, we first explore the special structure of **P0** to obtain an optimal solution to power allocation. Then, we further obtain our solution by utilizing BEM-FPA to directly solve **P0** with the given optimal power allocation.

We start with the following theorem:

Theorem 9. For low-SNR SGLs, the global optimal power to **P0** is the maximum transmit power, i.e., $P_i^* = P_{\max}^{s(i)}, \forall i$.

To prove Theorem 9, we require the following lemma:

Lemma 2. For any task i , the energy consumption E_i associated with either a low-SNR SGL or any ISL is independent of the transmit power P_i .

Proof. See Appendix I, available in the online supplemental material. $\square$

Proof of Theorem 9. See Appendix J, available in the online supplemental material. $\square$

Algorithm 2. Balanced Energy and Makespan (BEM)

Input: Set $\xi = \frac{1}{2}$, generation number N , population size U .
1: Produce an initial population $\mathcal{P} = \{\mathcal{P}_u, 1 \leq u \leq U\}$ in a random manner.
2: **while** $1 \leq n \leq N$ **do**
3: **for** $u = 1 : U$ **do**
4: Map $\mathcal{P}_u$ into a candidate solution $\mathcal{P}$.
5: **for** $i = 1 : |\mathcal{I}|$ **do**
6: **if** $h(s(i)) \in \mathcal{H}_1$ **then**
7: Let $P_i = P_{\max}^{s(i)}$.
8: **end if**
9: **end for**
10: Call BEM-FPA (i.e., Algorithm 1) to obtain the fitness value by using the new candidate solution $\mathcal{P}$.
11: Update $\mathcal{P}_u$.
12: **end for**
13: Execute the bio-inspired operations on $\mathcal{P}$ [43].
14: $n = n + 1$.
15: **end while**
16: Map the best chromosome into $\mathcal{P}$.
Output: $\mathbf{x}, \mathcal{P}$.

It follows from Theorem 9 that we can set $\mathbf{P}^* = \{P_{\max}^{s(i)}\}$ and equivalently transform **P0** into the following form

$$\begin{aligned} \mathbf{P12:} \min_{\mathbf{x}, \mathbf{P}^*} \mathcal{G} \sum_{i \in \mathcal{I}} \sum_{\substack{h \in \mathcal{H} \\ t \in \mathcal{F}(i, h)}} x_{iht} (t + p_{ih} - 1) \\ + \frac{1 - \lambda}{E_{\text{max}} - E_{\text{tol}}^*} E_{\text{con}} - \mathcal{U} \\ \text{s.t. (7) - (9),} \\ (11) - (13). \end{aligned} \quad (26)$$

Here

$$p_{ih} = \begin{cases} \frac{D_i N}{P_{\max}^{s(i)} G_{s(i)}^{\text{tran}} G_h^{\text{rec}} L_f L_l B_c \log 2e}, & h \in \mathcal{H}_1, \\ \frac{D_i \kappa T_s (E_b/N_0)_{\text{req}} M}{P_{\max}^{s(i)} G_{s(i)}^{\text{tran}} G_h^{\text{rec}} L_f L_l}, & h \in \mathcal{H}_2, \end{cases}$$

and

$$\begin{aligned} E_{\text{con}} &= \sum_{i \in \mathcal{I}_1} \frac{D_i N}{G_{s(i)}^{\text{tran}} G_h^{\text{rec}} L_f L_l B_c \log 2e} \\ &\quad + \sum_{i \in \mathcal{I}_2} \frac{D_i \kappa T_s (E_b/N_0)_{\text{req}} M}{G_{s(i)}^{\text{tran}} G_h^{\text{rec}} L_f L_l}. \end{aligned}$$

We can remove both the constant coefficient $\mathcal{G}$ and the constant term $\frac{1-\lambda}{E_{\max}-E_{\text{tol}}}E_{\text{con}} - \mathcal{U}$ from the objective function of **P12** without losing optimality. We observe that this has the same form as **P2**. As such, we can directly call BEM-FPA to solve **P12** to obtain a $\frac{1}{2}$ -approximate solution with given $P_i = P_{\max}^{s(i)}, \forall i$.

4.2.2 Heuristic Algorithm for the Scenarios with Moderate to High-SNR SGLs

For the scenarios with moderate to high-SNR SGLs, solving the problem of balancing the minimum total energy consumption and mean makespan by joint task scheduling and power control is more difficult. The main challenge is in finding an optimal power allocation for the SGLs. It is however observed that if the transmit power allocation to **P0** is fixed, it can be simplified to the scheduling problem of minimizing the mean makespan (i.e., **P0**^{fixed power}), which can be solved efficiently by using Algorithm 1. Thus, in the proposed solution, we equivalently transform **P0** into a nested problem with two levels of optimization. In the upper-level optimization, the master problem is to optimize transmit power by solving the following problem

$$\begin{aligned} \text{P13: } \min_{\mathbf{P}} \quad & g(\mathbf{P}) \\ \text{s.t. } \quad & p_{ih} = \frac{D_i}{R_{ih}^{\text{SGL}}}, \forall i \in \mathcal{I}, h \in \mathcal{H}_1, \\ & p_{ih} = \frac{D_i}{R_{ih}^{\text{ISL}}}, \forall i \in \mathcal{I}, h \in \mathcal{H}_2, \\ & P_i \in \mathcal{P}^{s(i)}, \forall i \in \mathcal{I}, \end{aligned}$$

where $\mathcal{P}^{s(i)} = \{P_k^{s(i)} | P_k^{s(i)} = P_{\min}^{s(i)} + k \frac{P_{\max}^{s(i)} - P_{\min}^{s(i)}}{|\mathcal{P}^{s(i)}|}, k = 0, 1, \dots, |\mathcal{P}^{s(i)}| - 1\}$ and $g(\mathbf{P})$ is the optimal objective value of **P0**^{fixed power} for a given $\mathbf{P}$. In the lower-level optimization, we optimize $\mathbf{x}$ by solving **P0**^{fixed power}.

We note that, by Lemma 2, the optimal power allocation for ISLs is still the maximum transmit power. As such, we rewrite **P13** in the following form

$$\begin{aligned} \text{P14: } \min_{\mathbf{P}} \quad & g(\mathbf{P}) \\ \text{s.t. } \quad & p_{ih} = \frac{D_i}{R_{ih}^{\text{SGL}}}, \forall i \in \mathcal{I}, h \in \mathcal{H}_1, \\ & p_{ih} = \frac{D_i}{R_{ih}^{\text{ISL}}}, \forall i \in \mathcal{I}, h \in \mathcal{H}_2, \\ & P_i = P_{\max}^{s(i)}, \forall i \in \mathcal{I}, h(s) \in \mathcal{H}_1, \\ & P_i \in \mathcal{P}^{s(i)}, \forall i \in \mathcal{I}, h(s) \in \mathcal{H}_2. \end{aligned}$$

A genetic framework is adopted to solve **P14**. First, we represent a candidate solution to **P14** (i.e., $\mathbf{P}$) as a chromosome. Then, we define $g(\mathbf{P})$ as the fitness function to evaluate each chromosome. Furthermore, we execute the bio-inspired operations involving mutation, crossover, and selection on these chromosomes. We omit these details here since they following a standard genetic framework [43]. In the following, we summarize the resultant BEM algorithm in Algorithm 2. The algorithm framework is illustrated in Fig. 4.

Next, we give Theorem 10 to state the complexity of the BEM algorithm.

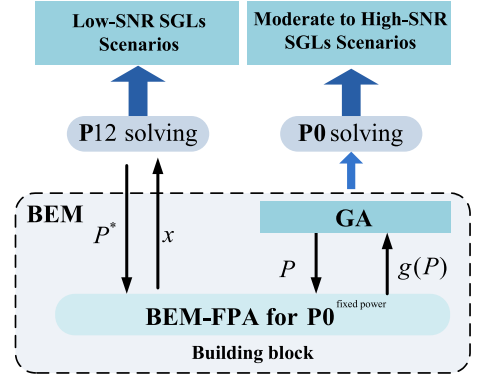


Fig. 4. Illustration for algorithm framework.

Theorem 10. The complexity of BEM is $\mathcal{O}(UN(IM)^{4.5} \log(1/\epsilon))$.

Proof. See Appendix K, available in the online supplemental material. $\square$

Finally, we analyse the convergence of two-layer algorithm BEM, which is composed of the outer-layer algorithm GA and the inner-layer algorithm BEM-FPA. In BEM-FPA, we utilize an interior-point method [44] to solve **P10**. From [44], it is easy to obtain that BEM-FPA is linear convergence with $\mathcal{O}(\log(1/\epsilon))$. Furthermore, to analyse the convergence of GA, we utilize the result of [45], which is based on the minorization condition in the Markov chain theory. From [45], we obtain that $\|\mu_\alpha - \pi\| \leq (1 - \delta)^{\lfloor \alpha/\alpha_0 \rfloor}$ where μ_α represents the probability distribution of population at iteration α , π is the probability distribution, δ denotes a positive constant with $0 < \delta < 1$, α_0 represents the first iteration, and $\lfloor \alpha/\alpha_0 \rfloor$ indicates the maximum integer less than or equal to α/α_0 . We let $(1 - \delta)^{\lfloor \alpha/\alpha_0 \rfloor - 1} \leq \epsilon$ to yield that $\lfloor \alpha/\alpha_0 \rfloor \geq C \log(1/\epsilon)$ with $C = \frac{1}{\log(1-\delta)} > 0$. Using the fact that $\lfloor \alpha/\alpha_0 \rfloor \leq \alpha/\alpha_0$, we have $\alpha \geq \alpha_0 C \log(1/\epsilon)$, which indicates GA is also linear convergence with $\mathcal{O}(\log(1/\epsilon))$. In brief, the convergence of BEM can be guaranteed, which is also verified in Fig. 7.

5 SIMULATION EVALUATION

In this section, we evaluate the performance of the proposed algorithms (i.e., BEM-FPA and BEM) through extensive simulation. We aim to verify the performance of the proposed algorithms for low-SNR scenarios and high-SNR scenarios in term of feasibility, convergence, and effectiveness. We make comparison with a lower bound of the optimum in the low-SNR case, and with non-dominated sorting genetic algorithm-II (NSGA-II) [46], random power allocation, and random antenna allocation in the high-SNR case.

5.1 Parameters Setting

A co-simulation platform composed of the satellite tool kit and MATLAB is used to conduct our simulation. The related parameters are listed in Table 2. We consider the GEO backbone of SAGIN involving three DRSs. Each DRS is equipped with four receiver antennas. Half of the tasks are offloaded on ISLs and the other half are on SGLs. The data amount of each task is randomly generated from a uniform distribution in [5,15] Gbits. The scheduling horizon is from 1 September 2021

TABLE 2
Simulation Parameters

Data sinks	Latitude	Longitude
DRS 1	0°	41°
DRS 2	0°	174°
DRS 3	0°	275°
L_f	10^{-23}	
$G_{s(i)}^{\text{tran}}$	36 dB	
$G_{h_1}^{\text{rec}}$	119.5894 dB	
$G_{h_2}^{\text{rec}}$	117.9284 dB	
$G_{h_3}^{\text{rec}}$	116.2675 dB	
$G_{h_4}^{\text{rec}}$	114.6065 dB	
Line loss of ISLs	-200 dB	
Line loss of SGLs	-210 dB	
$P_{\min}^{s(i)}$	100 watts	
$P_{\max}^{s(i)}$	200 watts	
T_s	316.2278 K	
L_l	1	
(E_b/N_0)	9.6 dB	
M	1.4125 dB	
κ	$1.38054 * 10^{-23} \text{ JK}^{-1}$	

00:00:00.0 to 2 September 2021 00:00:00.0. We solve **P10** optimally by using YALMIP [47] to call the SeDuMi solver [48].

The parameters of GA are listed as follows: The mutation probability is set to 0.8; the crossover probability is set to 0.5; the population size is set to 30; The number of genetic generations is set to 100.

The different settings for the low-SNR scenarios and high-SNR scenarios are listed as follows: In low-SNR scenarios, we set $L_f = 10^{-23}$ to satisfy $\text{SNR} \ll 1$. In high-SNR scenarios, L_f is set to 10^{-21} .

5.2 Performance Evaluation

5.2.1 Low-SNR Scenarios

In Figs. 5 and 6, we adopt the *lower bound* as a baseline to verify the performance gap between the optimum of **P0**^{fixed power} and the BEM-FPA algorithm. To find a lower bound of optimum, we use YALMIP to call the solver SeDuMi to solve the semidefinite programming problem **P10** optimally. We label the normalized weighted sum of total energy consumption and mean makespan as network cost.

In Fig. 5a, we compare BEM-FPA with the *lower bound* for different task number (TN) and antenna number (AN) in terms of the network cost versus λ . Furthermore, we obtain the total energy consumption and mean makespan from the network cost in Fig. 5a and further plot them versus λ in Figs. 5b and 5c.

From Fig. 5a, we observe that the network cost increases as λ increases. Meanwhile, from Figs. 5b and 5c, we observe that both total energy consumption and mean makespan remain constant with increasing λ . From Fig. 5b, we observe that total energy consumption remains constant with increasing λ . This corresponds with Lemma 2, i.e., the energy consumption for each task is not a function of its allocated transmit power in low-SNR scenarios. This illustrates that total energy consumption is fixed in low-SNR scenarios when both the number of tasks and the number of antennas are given. Moreover, Fig. 5b shows that total energy consumption increases as the number of antennas increases. This is because the gain of receiver antenna in the simulation is set to a decreasing value with

increasing antenna number. To minimize the mean makespan, more tasks are scheduled when there are more receiver antennas. Equation (29) indicates that energy consumption increases as the gain of receiver antenna decreases. Thus, the total energy consumption increases as the number of antennas increases.

Fig. 5c shows that mean makespan is unchange with increasing λ . This is because when total energy consumption is a fixed value in low-SNR scenarios, the studied problem degrades into a mean makespan minimization problem (i.e., **P1**). This means that there is no tradeoff between mean makespan and total energy consumption in low-SNR scenarios. Furthermore, it is observed from Fig. 5 that the performance gap between BEM-FPA and the *lower bound* is very small. In particular, the maximum performance gap to optimality is less than 2%, which suggests that BEM-FPA performs even better than what is predicted by the bound in Theorem 5. Since the optimum of **P12** lies between BEM-FPA and the *lower bound*, we see that BEM-FPA is nearly optimal.

Fig. 6a compares BEM-FPA with the *lower bound* for different λ and the number of antennas in terms of the network cost versus the number of tasks. Figs. 6b and 6c respectively plot total energy consumption and the mean makespan versus the number of tasks. We again observe from Fig. 6a that the performance of BEM-FPA is very close to that of the *lower bound* with the number of tasks increases. This further verifies the efficiency of BEM-FPA. From Fig. 6b, we observe that the total energy consumption increases as the number of tasks increase, which is expected. From Fig. 6c, the two designs with a smaller number of tasks or a larger number of antennas achieve a smaller mean makespan. This is because: when giving the number of antennas, the more tasks the larger their makespan; when giving the number of tasks, the tasks can be scheduled in advance on more antennas, thereby reducing their makespan. This phenomenon indicates that we can reduce the mean makespan by reducing task number or increasing antenna number, which is also expected.

5.2.2 High-SNR Scenarios

In Fig. 7, we show the convergence evolution of BEM with different values of λ in terms of network cost, total energy consumption, and mean makespan. It is observed from Fig. 7a that BEM typically converges in a small number of iterations. Fig. 7b shows that BEM obtains a larger total energy consumption with a larger value of λ as the number of iterations grows. In Fig. 7c, we observe that BEM achieves a smaller mean makespan with a larger value of λ as the number of iterations grows. The above observations reflect the tradeoff between the two objectives of minimizing total energy consumption and the makespan.

Fig. 8 shows the performance between BEM and various baseline schemes in terms of network cost, total energy consumption, and mean makespan. The three baseline schemes are NSGA-II, *random power*+BEM-FPA, and *random*. Specifically, under *random power*+BEM-FPA, we first allocate power $P_i \in \mathcal{P}^{s(i)}$ for any task i in a random fashion and then we run BEM-FPA given the random power allocation. Under *random*, both the power and transmit antenna are randomly allocated for each task.

Fig. 8a shows NSGA-II achieves a smaller network cost than that of *random*. This because NSGA-II using bio-inspired

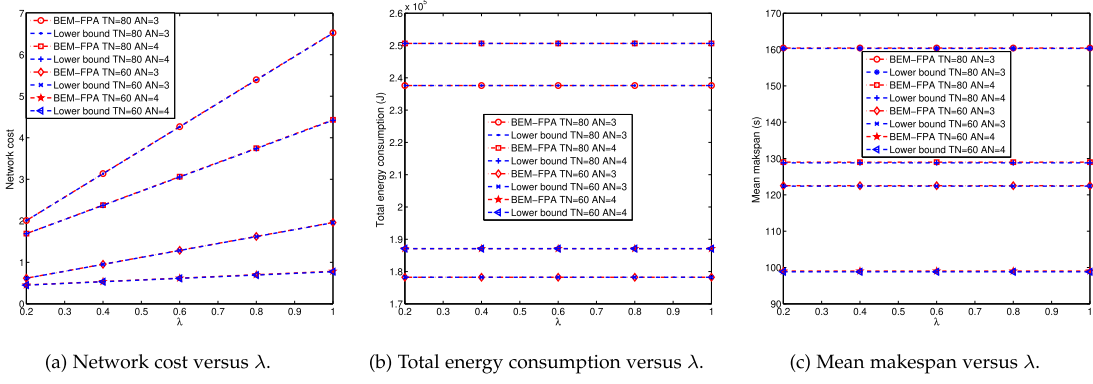
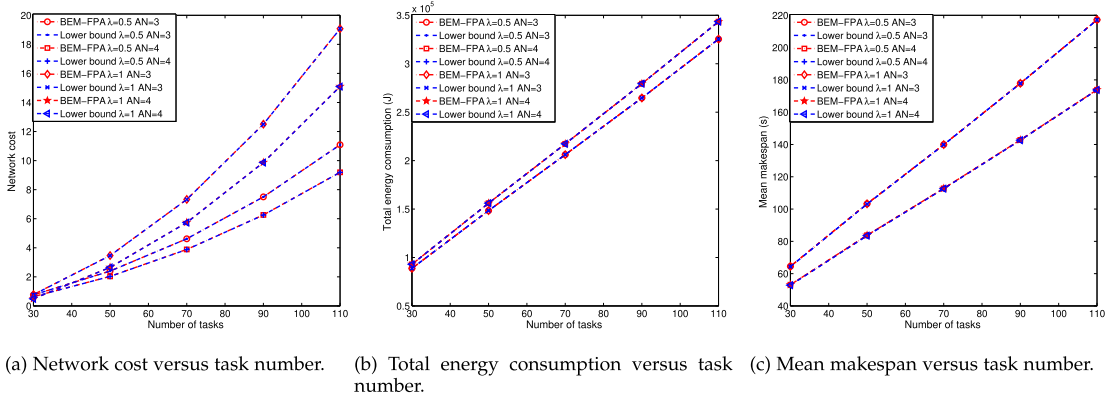
Fig. 5. Comparisons with lower bound of optimum versus λ .

Fig. 6. Comparisons with lower bound of optimum versus task number.

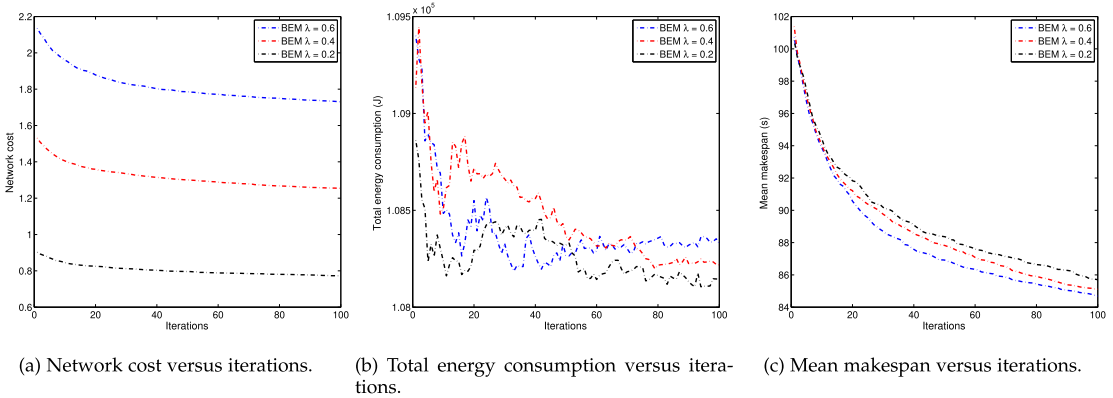


Fig. 7. Verification of convergence.

operations to yield some better solutions over multiple generations. One observes from Fig. 8a that the network cost obtained by *random power*+BEM-FPA is far less than that of NSGA-II. This is because our proposed BEM-FPA can achieve a smaller mean makespan than that of NSGA-II, which is reflected by Fig. 8c. This exhibits the efficiency of BEM-FPA. Furthermore, it is observed from Fig. 8a that BEM achieves the smallest network cost among all schemes. In particular, BEM can further decrease the network cost over *random power*+BEM-FPA although the performance gap between them is small. This indicates that optimizing power allocation is necessary for reducing the network cost. In Fig. 8b, we observe that BEM yields the smallest total energy consumption among all schemes over all λ values. Furthermore, Fig. 8c shows that BEM yields a smaller mean makespan than *random power*

+BEM-FPA. These demonstrate that the joint power control and task scheduling can not only decrease the total energy consumption but also reduce the mean makespan for SAGINs. In brief, by jointly optimizing power control and task scheduling, BEM can boost the performance of SAGINs.

6 CONCLUSION

In this paper, we have investigated the data offloading problem for SAGINs aiming to balance the total energy consumption and the mean makespan. We formulate this problem as a normalized weighted sum minimization problem under the practical constraints of data offloading and power control. We develop a constant-factor approximate algorithm termed BEM-FPA for the studied problem with given

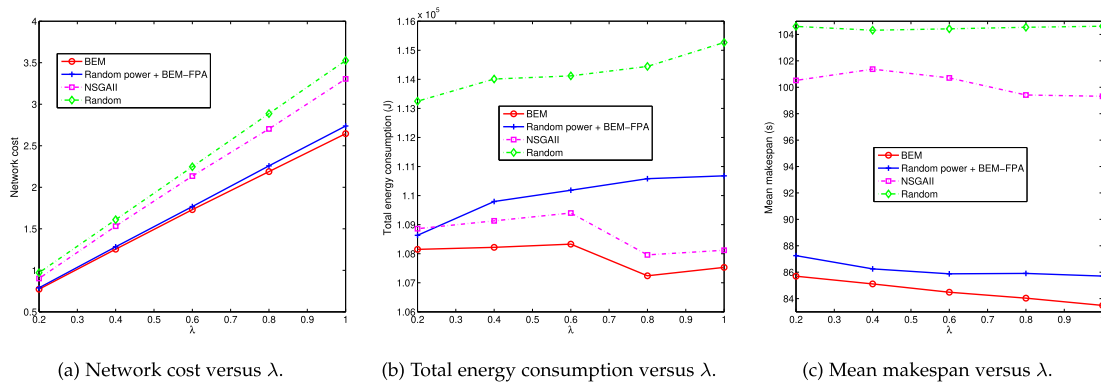


Fig. 8. Comparisons with alternatives.

power allocation, which is shown to be nearly optimal in simulation. We show that under low-SNR SGLs, BEM-FPA is directly applicable to the joint optimization problem, by allocating the maximum transmitting power for each task. Furthermore, to solve the problem for moderate to high-SNR SGLs, we devise a fast yet efficient data offloading algorithm termed BEM through combining BEM-FPA and a GA approach. Our simulation results exhibit that BEM is highly effective and superior to state-of-the-art solutions.

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